

Independent Derivation and Audit of the Citlali `convolve` Signal, Uncertainty, Response, and Support Contract

Citlali scientific-contract audit

2026-07-31

Abstract

This document derives the statistical contract for `post-processing.map-filter.type:convolve` independently of the candidate implementation. Its starting point is an affine estimator, not a particular C++ formula. The derivation separates stored filtered amplitude, response-corrected sky or template amplitude, statistical inverse variance, and dimensionless support or confidence. It then audits the baseline application and the two-commit correction candidate, identifies the remaining scientific choices, recommends a product contract, and specifies the evidence required before integration or fruit-loop use.

Contents

1	Executive summary	2
2	Scope, status labels, and terminology	3
2.1	Second-audit disposition	4
3	Notation	4
4	General affine estimator and covariance propagation	5
5	Interior fixed convolution	5
6	Validity, low formal weight, and variable coverage	6
7	Binary edge support and deterministic background fill	6
7.1	Support metrics	7
8	Locally normalized alternative	7
9	Fractional taper or apodization	8
10	Response normalization and photometric interpretation	9
11	Formal and empirical uncertainty products	9
12	Product-by-product recommended contract	10

13 Analytic limiting cases	12
13.1 Identity, boxcar, and uniform white noise	12
13.2 Nonuniform variance and a low-weight valid input	12
13.3 Constant sky and binary edge support	12
13.4 Delta and beam-shaped sources	13
13.5 Fractional tapers	13
13.6 Support with nonnegative and signed kernels	13
13.7 Central empirical $N = 2$ and rejection at $N = 1$	13
14 Implementation audit	13
14.1 Exact shared signal and edge operator	14
14.2 Baseline mainline	14
14.3 Candidate 1: <code>fa8f8a5b9</code>	15
14.4 Candidate successor and combined stack	15
14.5 Dormant fractional taper	16
14.6 Empirical and calibrated products	16
14.7 Kernel-square reproduction for retained TolTEC fixtures	17
14.8 Defect classification and verdict	17
15 Recommended Citlali contract	18
16 Consequences for fruit-loop feedback	19
17 Validation plan	20
17.1 Mathematical unit tests	20
17.2 Implementation and regression tests	20
17.3 Same-SHA Unity validation	20
17.4 Blank controls	20
17.5 Injection and recovery	20
18 Conclusions	21
A Implementation source references	21
B Product and test matrix	22
C Independent-derivation record	27

1 Executive summary

For a deterministic affine filter

$$\mathbf{y} = A\mathbf{x} + \mathbf{c}, \tag{1}$$

the stored filtered amplitude has covariance

$$C_y = AC_xA^\top. \tag{2}$$

For independent input pixels with formal variances V_j , its diagonal is

$$V_{y,i} = \sum_j A_{ij}^2 V_j. \tag{3}$$

Every input that enters the signal with a nonzero stochastic coefficient must therefore enter formal variance with the square of that same coefficient. Low positive input weight means high variance; it does not mean invalid support.

For a fixed, globally unit-sum convolution with a binary input edge window T_j , a binary output window O_i , and deterministic fill b_j , the independently derived stored product is

$$y_i = O_i \sum_j k_{ij} \{T_j x_j + (1 - T_j) b_j\}, \quad (4)$$

$$V_{y,i} = O_i^2 \sum_j k_{ij}^2 T_j^2 V_j, \quad (5)$$

$$W_{y,i} = V_{y,i}^{-1} \quad (V_{y,i} > 0 \text{ and valid}). \quad (6)$$

This is a fixed filtered amplitude, not a locally normalized sky estimator. The variance sum applies only to inputs with a defined stochastic variance. In the current candidate those are the finite positive-weight inputs; a zero or non-finite-weight input must carry no stochastic signal or noise contribution. Near an edge, its response and fill contribution must be exposed rather than hidden in a quantity called weight. A response-corrected estimator for a specified template p is a different product,

$$\hat{\alpha}_i = \frac{y_i - c_i}{R_i(p)}, \quad \text{Var}(\hat{\alpha}_i) = \frac{V_{y,i}}{R_i(p)^2}, \quad (7)$$

where $R_i(p) = \sum_j A_{ij} p_j$ and c_i is the deterministic affine fill.

The recommended Citlali contract is therefore the combination of options A, D, and E in the audit request: retain the fixed, globally normalized filtered amplitude and keep formal variance, empirical variance, response, support, and any confidence or feedback gain as distinct products. Until an explicit support equation and scientific threshold are chosen and validated, filtered products are withheld from production fruit-loop selection rather than being masked by an invented threshold. If photometric correction is required, add a separately named template-specific estimator rather than relabeling the fixed smoothing output as point-source flux.

2 Scope, status labels, and terminology

The document uses the following evidence labels.

Independent contract

A result derived from probability and estimator definitions before examination of commits `fa8f8a5b9` and `02a198cbf`.

Citlali approximation

A bounded simplification, principally diagonal input covariance and a stored variance plane rather than full output covariance.

Baseline

Behavior at canonical application mainline `9aae0e669384c5c0c0dda93debc194d6b8dac787`.

Candidate 1

Commit `fa8f8a5b939aa8e78f1b2fc0affc86857f2b2f9c`.

Candidate stack

Candidate 1 followed by `02a198cbfb379eaf6ab279c5a3d44ee73ff90435`.

Recommendation

The proposed contract; it is not accepted science until review and validation are complete.

Four terms are deliberately not interchangeable:

1. *stored filtered amplitude*: the numerical output of the filtering operator;
2. *response-corrected estimator*: an estimate of a named sky or template amplitude after correcting the realized filter response;
3. *statistical inverse variance*: the reciprocal variance of one explicitly named estimator; and
4. *support, validity, or confidence*: a mask, fractional geometric support, response guard, heuristic confidence, or feedback gain.

The unqualified word “weight” is avoided whenever one of these meanings can be stated instead.

2.1 Second-audit disposition

A second derivation was frozen before this document was opened, with SHA-256 digest

d2ad1ab7a8978e2c93c04a9e03585dd67e80bbdfd860c62797c47d253efb7710.

It substantively agreed with the numerical and product-contract conclusions. The cross-audit corrected the simplified edge equations to include the outer output window, made the zero-weight stochastic precondition explicit, qualified the $N = 1$ statement for the known-zero-mean estimator, corrected the exact `coverage_bool_I` comparison semantics and input-weight provenance, and clarified the non-periodic geometric dilation. Because the support-plane eligibility definition, coefficient norm, and scientific floor remain undecided, the bounded implementation chooses the permitted conservative outcome: no new support or response plane, and unit-sum filtered fruit-loop feedback is marked and enforced as withheld. The implementation persists the neutral signal-HDU producer identity `FILTEROP=unit_sum_convolution`; filtered loading rejects that identity, explicit `FLFBACK=false`, and missing producer identity. New full-Wiener products instead record `FILTEROP=wiener_filter`, preserving compatibility without claiming scientific approval. The comparison and action classification are recorded in `doc/CONVOLVE_SECOND_AUDIT_COMPARISON_2026-07-31.md`.

3 Notation

Input map pixels are x_j ; output pixels are y_i . Their expectations are μ_j and $\mu_{y,i}$. C_x and C_y are full covariance matrices. When the diagonal approximation is invoked, $V_j = (C_x)_{jj}$ and $V_{y,i} = (C_y)_{ii}$. A finite positive formal input inverse variance is $w_j > 0$ and

$$V_j = \frac{1}{w_j}. \quad (8)$$

k_{ij} is the fixed convolution coefficient from input j to output i . $T_j \in \{0,1\}$ is a binary input edge/operator window and $O_i \in \{0,1\}$ is a binary output window. Citlali currently realizes $O = T$. The corresponding diagonal matrices are also denoted T and O . T is not, by itself, the finite-positive-weight eligibility mask: the realized guardband can contain zero-weight pixels. t_j and a_i are fractional input and output tapers. b_j is an exterior fill. p_j is a specified sky template of unit amplitude and α its amplitude.

4 General affine estimator and covariance propagation

From Equation (1), for deterministic A and $\mathbf{c}$,

$$\mathbb{E}[\mathbf{y}] = A\mathbb{E}[\mathbf{x}] + \mathbf{c}, \quad (9)$$

$$C_y = \mathbb{E}[(\mathbf{y} - \mathbb{E}\mathbf{y})(\mathbf{y} - \mathbb{E}\mathbf{y})^\top] = AC_xA^\top. \quad (10)$$

Under $C_x = \text{diag}(V_j)$,

$$\text{Var}(y_i) = \sum_j A_{ij}^2 V_j, \quad (11)$$

$$\text{Cov}(y_i, y_\ell) = \sum_j A_{ij} A_{\ell j} V_j. \quad (12)$$

Equation (12) is essential: convolution correlates output pixels even when the input covariance is diagonal. A persisted variance plane is only $\text{diag}(C_y)$ and cannot encode the covariance needed for exact aperture, connected-region, or multi-pixel template significance.

If an exterior fill $\mathbf{b}$ is random rather than deterministic, write

$$\mathbf{y} = A\mathbf{x} + D\mathbf{b}, \quad (13)$$

where $A = OKT$ and $D = OK(I - T)$. Then

$$C_y = AC_xA^\top + DC_bD^\top + AC_{xb}D^\top + DC_{bx}A^\top. \quad (14)$$

Omitting the last three terms is justified only when the fill is fixed and known, or when its uncertainty and cross-covariance are demonstrably negligible. If $\mathbf{b} = H\mathbf{x} + \mathbf{d}$ is estimated linearly from the same map, the effective stochastic operator is $A + DH$ and the exact covariance is

$$C_y = (A + DH)C_x(A + DH)^\top. \quad (15)$$

5 Interior fixed convolution

For a fully supported fixed kernel,

$$y_i = \sum_j k_{ij} x_j, \quad (16)$$

so under diagonal input covariance

$$V_{y,i} = \sum_j k_{ij}^2 V_j, \quad (17)$$

$$\text{Cov}(y_i, y_\ell) = \sum_j k_{ij} k_{\ell j} V_j. \quad (18)$$

For translation-invariant k and uniform white input variance σ^2 ,

$$V_y = \sigma^2 \sum_j k_j^2. \quad (19)$$

If $\sum_j k_j = 1$, constant fields are preserved, while

$$\frac{\sigma_{\text{out}}}{\sigma_{\text{in}}} = \sqrt{\sum_j k_j^2}, \quad (20)$$

$$\frac{W_{\text{out}}}{W_{\text{in}}} = \frac{1}{\sum_j k_j^2}. \quad (21)$$

These statements concern the stored filtered amplitude. Unit DC gain does not imply unit response to a delta function, a beam template, or integrated point-source flux.

6 Validity, low formal weight, and variable coverage

Equation (8) gives the decisive distinction: $0 < w_j \ll 1$ is a valid, high-variance input. If the signal coefficient A_{ij} is nonzero, its variance contribution is A_{ij}^2/w_j . Retaining the sample in signal while omitting that term is a signal/variance operator mismatch and overstates significance.

If a policy threshold deliberately rejects a low-weight pixel, define $M_j = \mathbf{1}(w_j \geq w_{\min})$ and change the complete stochastic operator:

$$y_i = \sum_j A_{ij} M_j x_j + c_i, \quad (22)$$

$$V_{y,i} = \sum_j A_{ij}^2 M_j V_j, \quad (23)$$

$$R_i(p) = \sum_j A_{ij} M_j p_j. \quad (24)$$

Support must be computed from the same M_j . A threshold may instead define a displayed coverage diagnostic, but then it must not silently change only formal variance or only output validity.

7 Binary edge support and deterministic background fill

Define

$$x_j^* = T_j x_j + (1 - T_j) b_j \quad (25)$$

and apply the globally fixed kernel followed by the output window:

$$y_i = O_i \sum_j k_{ij} x_j^* = O_i \sum_j k_{ij} T_j x_j + c_i, \quad (26)$$

$$A_{ij} = O_i k_{ij} T_j, \quad (27)$$

$$c_i = O_i \sum_j k_{ij} (1 - T_j) b_j. \quad (28)$$

For deterministic fill and diagonal input covariance,

$$V_{y,i} = O_i^2 \sum_j k_{ij}^2 T_j^2 V_j, \quad (29)$$

$$\text{Cov}(y_i, y_\ell) = O_i O_\ell \sum_j k_{ij} k_{\ell j} T_j^2 V_j. \quad (30)$$

For a globally unit-sum kernel define the realized constant-sky response

$$r_i = O_i \sum_j k_{ij} T_j. \quad (31)$$

If the true sky and fill are constants s and b ,

$$\mathbb{E}[y_i] = r_i s + (O_i - r_i) b = O_i b + (s - b) r_i. \quad (32)$$

Thus zero fill attenuates a constant sky by r_i ; correct fill ($b = s$) preserves the output-windowed sky; and the edge bias relative to $O_i s$ is

$$\mathbb{E}[y_i] - O_i s = (O_i - r_i)(b - s). \quad (33)$$

For a compact template p , its stochastic response is

$$R_i(p) = O_i \sum_j k_{ij} T_j p_j. \quad (34)$$

Fill is an additive offset, not template response, and must be subtracted if a response-corrected template amplitude is formed.

Full support with $T_j = 1$ and $O_i = 1$ gives $c_i = 0$ and the interior equations. Partial support gives spatially varying response, variance, and covariance. One valid input q gives $y_i = O_i k_{iq} x_q + c_i$ and $V_{y,i} = O_i^2 k_{iq}^2 V_q$. Zero valid support gives deterministic $y_i = c_i$ and zero stochastic variance; an output with $O_i = 0$ is likewise masked. Either case is scientifically invalid, not an infinitely precise measurement. This proves that validity cannot be inferred from inverse variance alone.

7.1 Support metrics

For a nonnegative unit-sum kernel, r_i in Equation (31) is also a useful fractional kernel support. For a signed kernel, cancellation makes $\sum kT$ unsuitable as a geometric support metric. Two explicit alternatives are

$$q_i^{(|k|)} = \frac{O_i \sum_j |k_{ij}| T_j}{\sum_j |k_{ij}|}, \quad (35)$$

$$q_i^{(k^2)} = \frac{O_i^2 \sum_j k_{ij}^2 T_j^2}{\sum_j k_{ij}^2}. \quad (36)$$

$q^{(|k|)}$ measures retained absolute operator mass; $q^{(k^2)}$ measures retained variance leverage under uniform input variance. A binary-footprint fraction measures geometry but ignores coefficient importance. None is a statistical inverse variance. A future support plane must name its norm, input-eligibility mask, output mask, and validity floor and must keep signed template/DC response separate. The second audit found that the current edge window may include zero-weight guardband pixels, so neither expression is adopted as a production-selection product in the bounded repair. Filtered selection that requires such a product remains withheld pending the choice and validation of those semantics.

8 Locally normalized alternative

For $r_i = O_i \sum_j k_{ij} T_j \neq 0$, a locally normalized estimator is

$$y_i^{\text{LN}} = \frac{O_i \sum_j k_{ij} T_j x_j}{r_i}. \quad (37)$$

Its diagonal variance and covariance are

$$V_i^{\text{LN}} = \frac{O_i^2 \sum_j k_{ij}^2 T_j^2 V_j}{r_i^2}, \quad (38)$$

$$\text{Cov}(y_i^{\text{LN}}, y_\ell^{\text{LN}}) = \frac{O_i O_\ell \sum_j k_{ij} k_{\ell j} T_j^2 V_j}{r_i r_\ell}. \quad (39)$$

It preserves a constant sky where r_i is well conditioned, but changes the operator, noise, compact-source response, and covariance with position. Near small or cancellation-dominated r_i , its variance can diverge.

On an accepted output with $O_i = 1$ and binary T , the expression

$$\frac{\sum_j k_{ij}^2 T_j V_j}{\sum_j k_{ij}^2 T_j} \quad (40)$$

is not the variance of Equation (37): its denominator lacks the square of $\sum kT$. It would equal the variance of an estimator with coefficients $kT/\sqrt{\sum k^2 T}$, but only if the signal were normalized by that same square-root factor. Dividing only variance by k^2 support therefore has no matching signal estimator and is a definite operator mismatch.

9 Fractional taper or apodization

For deterministic input taper t_j and output taper a_i ,

$$y_i = a_i \sum_j k_{ij} t_j x_j, \quad (41)$$

$$A_{ij} = a_i k_{ij} t_j, \quad (42)$$

$$V_{y,i} = a_i^2 \sum_j k_{ij}^2 t_j^2 V_j, \quad (43)$$

$$W_{y,i} = \frac{1}{a_i^2 \sum_j k_{ij}^2 t_j^2 V_j} \quad \text{when valid.} \quad (44)$$

The powers are fixed by the signal coefficients: input taper is squared inside the sum and output taper is squared outside. Applying a taper once in signal but once, twice, or with a reciprocal heuristic in weight space cannot be formal inverse variance unless it reduces algebraically to Equation (44).

For template p ,

$$R_i^{(t,a)}(p) = a_i \sum_j k_{ij} t_j p_j, \quad (45)$$

and the response-corrected variance is

$$\text{Var}(\hat{\alpha}_i) = \frac{a_i^2 \sum_j k_{ij}^2 t_j^2 V_j}{[a_i \sum_j k_{ij} t_j p_j]^2}. \quad (46)$$

For nonzero a_i , it cancels. Deterministic output taper scales amplitude and uncertainty together and creates no information. Input taper changes the relative coefficients and generally does not cancel. Tapering, discarding samples, locally renormalizing a truncated kernel, masking an output, and multiplying by heuristic confidence are five distinct operations.

Until a single estimator, response product, and formal variance with the exact powers above are specified and tested, fractional taper should remain rejected for accepted `convolve` configurations. A redesigned future mode should store taper, response, and support independently rather than encoding them in formal inverse variance.

10 Response normalization and photometric interpretation

For $\mathbb{E}[x_j] = \alpha p_j + \mu_{0,j}$, define

$$R_i(p) = \sum_j A_{ij} p_j. \quad (47)$$

After subtracting the deterministic affine offset and any named baseline response, a local estimator is

$$\hat{\alpha}_i = \frac{y_i - c_i - (A\boldsymbol{\mu}_0)_i}{R_i(p)}, \quad (48)$$

$$\text{Var}(\hat{\alpha}_i) = \frac{V_{y,i}}{R_i(p)^2}. \quad (49)$$

This is meaningful only when p , its alignment, the local baseline, and a well-conditioned nonzero response are defined. When the template occupies multiple correlated output pixels, the appropriate generalized least-squares estimator is

$$\hat{\alpha} = \frac{\mathbf{r}^\top C_y^{-1}(\mathbf{y} - \mathbf{c} - \boldsymbol{\mu}_{y,0})}{\mathbf{r}^\top C_y^{-1} \mathbf{r}}, \quad (50)$$

$$\text{Var}(\hat{\alpha}) = \frac{1}{\mathbf{r}^\top C_y^{-1} \mathbf{r}}, \quad (51)$$

where $\mathbf{r} = A\mathbf{p}$. A diagonal variance plane alone is insufficient for the exact form.

Response identities must name the template:

- DC normalization is $R(\mathbf{1})$; a unit-sum interior kernel gives one.
- Delta peak response is the appropriate kernel coefficient, commonly k_0 , not $\sum k$.
- A beam-template peak response is $(k * p)$ evaluated at the chosen center.
- Integrated response depends on pixel solid angle, map unit convention, and aperture or template normalization.
- Matched-filter normalization is set by the template and noise covariance, not by unit DC gain.

Thus **mJy/beam** amplitude and **mJy** integrated flux are not interchangeable. A unit-sum smoother preserves a constant field in the interior but generally lowers the peak of a point source.

For any nonzero deterministic response correction R ,

$$\frac{y/R}{\sqrt{V_y/R^2}} = \frac{y}{\sqrt{V_y}} \quad (52)$$

up to the sign convention for R . Correcting both amplitude and uncertainty does not improve S/N.

11 Formal and empirical uncertainty products

Within the diagonal approximation, a formal inverse variance must be derived from one named variance:

$$W_i^{\text{formal}} = \begin{cases} 1/V_{y,i}, & V_{y,i} > 0 \text{ finite and output valid,} \\ \text{invalid,} & \text{otherwise.} \end{cases} \quad (53)$$

Then

$$\sigma_i^{\text{formal}} = \sqrt{V_{y,i}}, \quad (54)$$

$$Z_i^{\text{formal}} = \frac{y_i}{\sqrt{V_{y,i}}}. \quad (55)$$

The latter is a formal standardized amplitude, not automatically an empirically calibrated significance.

For $N \geq 2$ jackknife realizations $z_i^{(n)}$ filtered with the same operator, fixed support, fixed fill, and fixed weights,

$$\bar{z}_i = \frac{1}{N} \sum_{n=1}^N z_i^{(n)}, \quad (56)$$

$$s_i^2 = \frac{1}{N-1} \sum_{n=1}^N (z_i^{(n)} - \bar{z}_i)^2. \quad (57)$$

This is a diagonal empirical variance conditional on the realized filtering policy. It does not capture coherent artifacts common to every realization, and spatial covariance remains unrepresented. If realization masks, weights, support, fill, or normalization change, the $z^{(n)}$ are not samples of one common estimator without further modeling.

A globally calibrated formal spatial pattern may be defined explicitly as

$$\gamma = \text{Calibrate}\left(\{s_i^2, V_{y,i}^{\text{formal}}\}_{i \in \mathcal{Q}}\right), \quad (58)$$

$$V_i^{\text{calibrated}} = \gamma V_{y,i}^{\text{formal}}, \quad W_i^{\text{calibrated}} = \frac{W_i^{\text{formal}}}{\gamma}. \quad (59)$$

It is not pixelwise $1/s_i^2$ unless that is explicitly the chosen definition. The product name and metadata must state which pattern is formal, which scalar is empirical, the calibration region $\mathcal{Q}$, and the estimator to which it applies.

12 Product-by-product recommended contract

Table 1 is the implementation-independent target. Exact baseline and candidate extension names, producers, metadata, and test dispositions are added in the implementation audit.

Table 1: Recommended product meanings.

Product concept	Units	Defining equation	Contract
Stored signal	map unit	(26)	Fixed globally normalized filtered amplitude, conditional on the realized window and data-derived fill where applicable.
Formal variance	map unit ²	(29)	Diagonal variance of the stored amplitude under the stated input-covariance approximation.

Product concept	Units	Defining equation	Contract
Formal inverse variance	map unit ⁻²	(53)	Reciprocal formal variance only where the conditional estimator is numerically valid; it is not support or confidence.
Empirical variance	map unit ²	(57)	$N - 1$ central sample variance of identically filtered realizations; $N \geq 2$.
Calibrated inverse variance	map unit ⁻²	(59)	Formal spatial pattern times a documented empirical calibration, if that policy is chosen.
Stored-amplitude S/N	dimensionless	(55) or empirical analog	Name whether formal-standardized or empirical.
Template response	dimensionless under matching template convention	(47)	Possible future template-specific signed response, separate from support; no authoritative convolve template response is currently persisted.
Absolute support	dimensionless	(35)	Possible future fractional retained absolute operator mass; the norm, eligibility mask, and floor remain unapproved. Never inverse variance.
Coverage	seconds	producer exposure equation	Exposure diagnostic; a <code>cov_cut</code> display threshold must not alter only variance.
Kernel	current kernel map unit by repository convention	(16)	Realized fixed kernel plus normalization and template identity metadata.
Point-source amplitude alias	map unit	(26)	Compatibility alias only unless a beam/template response correction is actually applied.
Point-source uncertainty alias	map unit	named formal or empirical stored-amplitude variance	Square root of a named variance of the same aliased amplitude; the current convolve alias uses direct empirical sample variance.
Point-source S/N alias	dimensionless	(52)	Ratio of the two compatible aliases; not absolute photometry.

Product concept	Units	Defining equation	Contract
Confidence/gain	dimensionless	policy-specific	Separate optional heuristic; cannot be named weight or variance.

Invalid output requires an explicit validity authority. Zero stochastic variance at zero support is invalid, not infinite inverse variance. If a response or support product is introduced later, it must be an explicit dimensionless plane: response must identify its template, while support must identify its eligibility mask, coefficient norm, and validity floor.

13 Analytic limiting cases

All cases below follow from the preceding equations and were written before inspection of the candidate implementation.

13.1 Identity, boxcar, and uniform white noise

For $k = (1)$ and $O = T = 1$, $y = x$, $V_y = V$, $W_y = w$, response and support are one. For an m -pixel boxcar $k_j = 1/m$ with full support and uniform variance σ^2 ,

$$y = \frac{1}{m} \sum_j x_j, \quad V_y = \frac{\sigma^2}{m}, \quad W_y = mW_{\text{in}}. \quad (60)$$

For $m = 4$, $\sum k^2 = 1/4$, so the output RMS is one half the input RMS and the inverse variance is four times the input inverse variance.

13.2 Nonuniform variance and a low-weight valid input

For $k = (1/2, 1/2)$, $V_1 = 1$, and $V_2 = 100$,

$$y = \frac{x_1 + x_2}{2}, \quad V_y = \frac{1}{4}(1 + 100) = 25.25, \quad \sigma_y \simeq 5.0249, \quad W_y \simeq 0.039604. \quad (61)$$

The second input has formal weight 0.01 but remains valid. Omitting it only from variance would report $V_y = 0.25$, understate uncertainty by a factor $\sqrt{101} \simeq 10.05$, and overstate $|S/N|$ by the same factor while the signal still depends equally on x_2 .

13.3 Constant sky and binary edge support

Take $k = (1/2, 1/2)$, true constant sky $s = 10$, output window $O = 1$, and $T = (1, 0)$, so $r = 1/2$. With zero fill, $y = 5$, $c = 0$, and the fixed-amplitude response to the constant sky is $1/2$. With fill $b = 4$, $y = 7$, $c = 2$, and Equation (32) gives $4 + (10 - 4)/2 = 7$. With correct fill $b = s = 10$, $y = 10$ despite partial stochastic support. If the retained input variance is $V_1 = 4$, formal variance is $V_y = (1/2)^2 4 = 1$ in all three deterministic-fill cases. The response-corrected constant-sky estimate is $(y - c)/r = 10$ with variance $1/r^2 = 4$.

Full support $T = (1, 1)$ gives $r = 1$ and the interior estimator. One valid contributor gives the result above. Zero support $T = (0, 0)$ gives only the fill contribution and zero stochastic variance but must be marked invalid.

13.4 Delta and beam-shaped sources

For a unit delta at input q and zero fill,

$$y_i = O_i k_{iq} T_q, \quad R_i(\delta_q) = O_i k_{iq} T_q. \quad (62)$$

The peak is k_{qq} for a centered translation-invariant kernel, not one in general. Under full support $O = T = 1$, a unit-amplitude beam template p has output $k * p$ and central peak response $R_0(p) = \sum_j k_{0j} p_j$. A beam-shaped source is therefore broadened and its peak generally changed even though DC response is one. Integrated-flux preservation additionally depends on map units, pixel solid angle, beam convention, and aperture; it does not follow from the peak or DC response alone.

13.5 Fractional tapers

With $k = (1/2, 1/2)$, $t = (1, 1/2)$, $a = 1$, and equal variance $V = 4$,

$$y = \frac{1}{2}x_1 + \frac{1}{4}x_2, \quad V_y = 4 \left(\frac{1}{4} + \frac{1}{16} \right) = 1.25. \quad (63)$$

An output taper $a = 1/2$ changes this to $y/2$ and $V_y/4 = 0.3125$; formal inverse variance rises from 0.8 to 3.2 solely because the stored amplitude unit scale changed. The response-corrected estimator and its information are unchanged because the output taper cancels as in Equation (46). Applying both tapers uses both stated powers; it is not a sample-count correction.

13.6 Support with nonnegative and signed kernels

For output window $O = 1$, nonnegative $k = (1/4, 1/2, 1/4)$, and $T = (1, 1, 0)$, $r = q^{(|k|)} = 3/4$, while $q^{(k^2)} = (1/16 + 1/4)/(3/8) = 5/6$. They answer different questions. For signed $k = (-1, 1, 1)$ normalized to sum one and $T = (1, 1, 0)$, the signed mask convolution is zero even though two thirds of the footprint and two thirds of the absolute operator mass are present: $q^{(|k|)} = 2/3$. Signed response zero is real for a constant template, but it is not zero geometric support.

13.7 Central empirical $N = 2$ and rejection at $N = 1$

For filtered realizations $z^{(1)} = 2$ and $z^{(2)} = 6$, $\bar{z} = 4$ and

$$s^2 = \frac{(-2)^2 + 2^2}{2 - 1} = 8. \quad (64)$$

Any stable online central-moment accumulation must return exactly this value within arithmetic precision. At $N = 1$, the mean-subtracted central sample variance in Equation (57) is undefined; any product that requests that estimator must fail closed rather than emit zero, infinity, or a formal substitute under an empirical label. The separately named known-zero-mean second-moment estimator remains defined for one realization.

14 Implementation audit

This section was written only after the independent draft through Section 13 was complete and its SHA-256 digest had been recorded as

c7c10e902bcc0f5355af513ccc8782dbbf2b0bb63fdfd0828191b3e32f78c28e.

Line references identify the stated Git object, because later source movement must not change the evidence being audited.

14.1 Exact shared signal and edge operator

Both baseline and candidate use an FFT convolution, hence a periodic/circular discrete operator K on the finite map. The filter template is required to have a sufficiently conditioned nonzero sum and is divided by that sum. Let h denote this globally unit-sum kernel. The edge science region is selected from positive formal weight above the `cov_cut`-derived threshold and, when available, positive coverage above a fraction of its core median. The region is non-periodically dilated by a Euclidean disk clipped at the map boundary, with radius

$$d = \max\{1, \lceil f_{\text{guard}} \max(\text{FWHM}_{\text{pix}}, 1) \rceil\} \quad (65)$$

to create the accepted binary window T . Thus a pixel below `cov_cut` may still have $T = 1$ because it lies in the guardband.

The exterior fill is not a configured constant. It is

$$b(\mathbf{x}) = \text{median}\{x_j : j \text{ is finite and in the science region}\}, \quad (66)$$

with zero as the empty-set fallback. The exact stored signal is

$$\mathbf{y} = D_T K \{D_T \mathbf{x} + (I - D_T)b(\mathbf{x})\mathbf{1}\}, \quad (67)$$

where $D_T = \text{diag}(T)$. The final D_T is an output mask applied after convolution. The mapmaking kernel image is pre- and post-multiplied in the same manner, but it is a data product being filtered, not the convolution template h itself.

Equation (67) is nonlinear because the median and the weight/coverage-derived window depend on realized data products. Conditional on the realized T and b , its stochastic signal operator is

$$A_{\text{cond}} = D_T K D_T, \quad \mathbf{c}_{\text{cond}} = D_T K (I - D_T)b\mathbf{1}. \quad (68)$$

The candidate formal calculation treats both as fixed. It therefore omits the fill covariance and cross-covariance in Equation (14), and also conditions on the realized window. This may be a defensible approximation after measurement, but it is not an unconditional formal covariance and is currently undocumented and unvalidated.

Noise realizations use

$$\mathbf{z}_{\text{out}}^{(n)} = D_T K D_T \mathbf{z}_{\text{in}}^{(n)}. \quad (69)$$

They reuse science-derived T and do not re-estimate or add b . They therefore sample the conditional stochastic operator in Equation (68), but not median-fill uncertainty, realization-dependent masks, or artifacts common to all jackknives.

14.2 Baseline mainline

Define $M_j = 1$ only where the post-edge input weight is finite, positive, and at least the `cov_cut`-derived threshold, and zero otherwise. Baseline forms two circular convolutions with h^2 and reports, on output-window support,

$$\tilde{V}_i = \sum_j h_{ij}^2 M_j V_j, \quad (70)$$

$$m_i = \sum_j h_{ij}^2 M_j, \quad (71)$$

$$W_i^{\text{base}} = \frac{m_i}{\tilde{V}_i} T_i^2. \quad (72)$$

Its implied variance is Equation (40). The signal in Equation (67) is divided by neither m_i nor $\sqrt{m_i}$, and guardband inputs below `cov_cut` still enter signal. Baseline therefore has two definite mathematical defects: the k^2 -support division has no matching stored signal estimator, and a low positive-weight input can enter signal while being omitted from formal variance. In a uniform, fully supported interior, the first defect cancels the expected variance reduction and leaves formal inverse variance equal to its input value instead of Equation (21).

`cov_cut` consequently has several distinct roles: it seeds the edge science mask; it defines the final displayed `coverage_bool_I` from the current output weight; it selects pixels for empirical global calibration; and, only in baseline and Candidate 1, it incorrectly removes variance terms without changing the signal. It is not a single scientific support definition. The exact C++ display-mask predicate is the negated strict comparison $\neg(W_i < \tau_{mcov})$, not a finite-positive validity test; in particular, a NaN comparison is false and therefore produces mask value one.

14.3 Candidate 1: fa8f8a5b9

Candidate 1 removes the division by m_i and intends to implement Equation (29), adds a relative validity floor, requires $N \geq 2$ for central empirical variance, applies the $N/(N-1)$ correction to its accumulated central moment, and changes the point-source compatibility metadata. However, it retains M_j 's `cov_cut` criterion while the signal retains all positive-weight inputs allowed by T . It therefore implements

$$V_i^{(1)} = T_i^2 \sum_j h_{ij}^2 M_j V_j, \quad (73)$$

which remains a signal/variance operator mismatch. Its difference-of-moments accumulation is also unnecessarily vulnerable to cancellation for large offsets. The commit is a useful but internally incomplete correction.

14.4 Candidate successor and combined stack

Commit 02a198cbf removes `cov_cut` from the input variance mask, uses every finite positive input weight that the binary window passes, and replaces difference-of-moments accumulation with Welford's stable central moment. For the accepted binary mode it implements

$$V_i^{\text{cand}} = T_i^2 \sum_{j:0 < w_j < \infty} h_{ij}^2 T_j V_j, \quad (74)$$

$$q_i^{\text{valid}} = \frac{\sum_{j:0 < w_j < \infty} h_{ij}^2 T_j}{\sum_j h_{ij}^2}, \quad (75)$$

$$W_i^{\text{cand}} = \begin{cases} 1/V_i^{\text{cand}}, & q_i^{\text{valid}} > 10^{-6}, V_i^{\text{cand}} > 0 \text{ finite}, T_i = 1, \\ 0, & \text{otherwise.} \end{cases} \quad (76)$$

The floor is a dimensionless fraction of total h^2 mass and remains valid for signed kernels. It controls only whether the formal output weight is zero; it neither renormalizes signal nor variance and is not persisted as a support plane. Conditional on T and b , and provided every zero or non-finite-weight input carries no unmodelled stochastic contribution, Equations (74) and (76) are the correct diagonal variance and inverse variance of the stored amplitude in Equation (67). Normal application `MapBuffer::normalize_maps` preparation zeros signal and every noise realization where effective weight is zero; lower-level callers must preserve that precondition. The unresolved data-derived-fill covariance remains.

The successor also rejects the dormant cosine taper for accepted `convolve` and Wiener `lowpass_only`; fixes output-policy routing so the effective typed configuration, rather than a product-output side effect, controls empirical normalization; records `NNOISE`, `VAREST=central_sample_variance`, `VARDDOF=1`, and `MEANSUB=true`; and makes `convolve` point-source aliases `TYPE=convolved_amplitude` without `RESPNORM`. It tests exact $N = 2$, mean-subtracted $N = 1$ rejection, and large-offset stability. The known-zero-mean second-moment path remains a separate estimator.

The load-time guard uses producer metadata rather than the consumer run's current filter configuration. Historical filtered products missing both `FILTEROP` and explicit `FLFBACK` approval fail closed because their producer operator cannot be established from the signal HDU. This is a restart/product-provenance guard, not a new response or support estimator.

The serial and OpenMP implementations follow the same intended sequence. Existing focused and regression gates establish code-path consistency, not scientific acceptance. In particular, they do not establish median-fill negligibility, edge response, covariance, S/N calibration, or astronomical recovery.

14.5 Dormant fractional taper

The dormant path replaces x by $b + (x - b)t$, multiplies input weight by t^2 , convolves, and then multiplies output signal by a and output weight by a^2 . Formal propagation begins from the already tapered weight, hence uses $1/(wt^2) = V/t^2$ inside the h^2 convolution, followed by multiplication of inverse variance by a^2 . These are the reciprocal of the required input power and the wrong output power for Equation (43). It is a definite signal/variance mismatch for stored tapered amplitude. Rejecting this path for the accepted modes is correct; redesign would require a new explicit estimator rather than a small weight-space patch.

14.6 Empirical and calibrated products

For central empirical products the successor computes Equation (57) with Welford accumulation. On calibration pixels $\mathcal{Q}$ selected by positive formal weight, positive empirical variance, and the `cov_cut` threshold, it then computes

$$\rho = \text{median}_{i \in \mathcal{Q}} \left(W_i^{\text{formal}} s_i^2 \right), \quad (77)$$

$$g = \frac{1}{\rho}, \quad (78)$$

$$W_i^{\text{empirical}} = g W_i^{\text{formal}}. \quad (79)$$

When empirical normalization is enabled, `weight_I` is replaced by Equation (79). It is a formal spatial pattern with one empirical scalar, not pixelwise $1/s_i^2$, and generally

$$W_i^{\text{empirical}} \neq \frac{1}{s_i^2}. \quad (80)$$

Here W^{formal} names the snapshot entering this filtered empirical calibration stage. It is not necessarily a pristine mapmaker weight: the snapshot can inherit an upstream scalar empirical calibration. Its metadata must therefore identify both the propagated estimator and the input-weight

provenance. The two empirical ratios written by the candidate are consequently

$$Z_i^{\text{pixel}} = y_i \sqrt{W_i^{\text{empirical}}}, \quad (81)$$

$$\sigma_i^{\text{direct}} = \sqrt{s_i^2}, \quad (82)$$

$$Z_i^{\text{direct}} = \begin{cases} y_i / \sigma_i^{\text{direct}}, & s_i^2 > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (83)$$

They are different estimators: the first retains the formal spatial pattern and a robust global scale, while the second uses each pixel’s direct sample variance. Neither represents a response-corrected point-source flux.

14.7 Kernel-square reproduction for retained TolTEC fixtures

The accepted science artifact at source SHA `a7a35a00` contains raw kernel maps for all three arrays. An independent checker reconstructed the convolution template using the baseline radial-binning and spline algorithm, with the retained WCS (405 by 465 pixels, 2 arcsec per pixel) and `kernel_template_tail_mode=constant`. No candidate value was used. Table 2 reports double-precision sums after global unit-sum normalization.

Table 2: Independently reproduced accepted-fixture kernel norms.

Array	raw $\sum h$	raw $\sum h $	normalized $\sum h^2$	$W_{\text{out}}/W_{\text{in}}$
a1100	4.189623968	19.728602743	0.288086271	3.471182426
a1400	-3.018860815	32.472567725	0.706692773	1.415042063
a2000	16.106367298	36.406942802	0.047908208	20.873249926

The raw a1400 template is signed and has negative DC sum; global division still gives unit DC gain, but the small ratio $|\sum h| / \sum |h|$ (0.093) demonstrates strong cancellation and reinforces the need to separate signed response from absolute support. These values verify Equations (20) and (21) for the reconstructed fixture; a same-SHA reduction must confirm the realized runtime template before acceptance.

14.8 Defect classification and verdict

The audit classifications are:

1. **Definite mathematical error:** baseline divides propagated variance by local h^2 support without applying the corresponding square-root normalization to signal (Equation (72)).
2. **Signal/variance operator mismatch:** baseline and Candidate 1 omit below-cov_cut variance contributions that remain in signal.
3. **Signal/variance operator mismatch:** dormant fractional taper uses reciprocal input-taper variance powers and the wrong output-weight power.
4. **Product naming/metadata error:** baseline labels unnormalized convolved-amplitude aliases as point-source products and claims `RESPNORM=1`; the successor corrects this.
5. **Diagonal-covariance approximation:** every variance plane omits Equation (12); no output covariance product exists.

6. **Unresolved approximation:** b is a nonlinear same-map median, but its variance and cross-covariance are absent; T is also treated as fixed.
7. **Inverse-variance versus support risk:** the h^2 validity floor, edge window, and `coverage_bool_I` are distinct policies, but no explicit edge-support or response plane is persisted.
8. **Scientifically defensible convention:** fixed globally normalized amplitude, conditional diagonal propagation, binary masking, and global empirical calibration are coherent when named honestly and validated.

The two-commit stack should be **amended:** retain its numerical core as one coherent conditional stored-amplitude correction, with the successor superseding Candidate 1’s low-weight and accumulation details, but amend the contract, metadata, support/response product decision, and validation evidence before integration. The commits need not be split on scientific grounds; neither should be integrated unchanged, and no integration is performed here.

15 Recommended Citlali contract

The recommended estimator is the fixed affine operator in Equations (26)–(30). Any future response or support products must remain distinct; the bounded repair withholds filtered production selection while their identities are unresolved. Specifically:

1. Normalize a scientifically admissible smoothing kernel globally and record its normalization and sign/conditioning policy.
2. Apply the realized binary input and output windows consistently to signal, formal variance, response, empirical realizations, and any future support product. A valid low-weight input remains in both signal and variance. A zero or non-finite-weight input is conditioned on as carrying no stochastic contribution; normal application preparation enforces this by zeroing its signal and noise, and lower-level callers must do the same.
3. Treat background fill as a named deterministic affine contribution. If it is data-derived, either propagate Equation (14) or state and validate the omitted uncertainty approximation.
4. Store the fixed filtered amplitude, its formal variance/inverse variance, the $N - 1$ empirical variance when $N \geq 2$, and any calibrated weight with truthful estimator metadata.
5. Do not claim or store a response without a named template. Do not promote the internal h^2 numerical-overlap guard, the edge window, or `coverage_bool_I` to scientific support. Before production selection, choose and validate a dimensionless support equation, eligibility mask, and floor; until then mark unit-sum-convolved products as selection-unready.
6. Retain historical point-source extension names only as explicitly marked `convolved_amplitude` compatibility aliases. Add actual photometric products only with a named beam/template, response correction, units, and validation.
7. Continue rejecting fractional taper in accepted `convolve` and Wiener `lowpass_only` configurations until it is redesigned around Equations (41)–(46).

This chooses fixed behavior over silent local normalization. A future local estimator may be scientifically useful, but it must be a separate named mode implementing both Equations (37) and (38), with response and conditioning gates.

The review must still decide (i) whether the same-map median-fill covariance is propagated, bounded as negligible, or replaced by a deterministic policy; (ii) the scientific template and normalization of any response plane; (iii) whether $q^{(|k|)}$, $q^{(k^2)}$, or both are exposed and what evidence sets their validity floors; (iv) whether empirical calibration remains a single robust scale or becomes a modeled spatial correction; (v) the intended photometric interpretation of `mJy/beam` after smoothing; and (vi) how filtered covariance is represented or supplied to downstream multi-pixel analyses. Those are product/science decisions, not implementation details.

16 Consequences for fruit-loop feedback

Pending an approved support/response contract and external validation, the frozen fruit-loop spatial estimator must remain fail-closed. No filtered product may currently select feedback support or supply feedback amplitude, weight, or gain. The inspected draft consumer correctly proposes raw coadd amplitude for feedback and uses filtered maps only to select spatial support, but it currently relies on `coverage_bool_I`, which is a final-weight threshold mask rather than an edge-response or absolute-support plane. Under the recommended contract:

- Future credible support may use an explicit validity mask plus an approved explicit support plane; neither is amplitude or inverse variance.
- Local thresholding may use an empirically calibrated stored-amplitude S/N only where its operator, support, fill, and covariance limitations are accepted. Formal-standardized amplitude is not a substitute under an empirical S/N label.
- Morphological selection may consume a threshold mask but must calibrate connected-component sizes and dilation under the filter-induced covariance of Equation (12).
- Feedback weighting and gain are separate policy quantities. Neither formal inverse variance nor support automatically defines a stable feedback gain.
- The amplitude projected into the timestream should remain the explicitly chosen raw coadd amplitude unless a separately reviewed filtered/response- corrected amplitude contract is adopted. Detection support must not silently replace feedback amplitude.

Only after the required validation and approval of an explicit support and response policy could `sig2noise_point_source_I`—despite its legacy extension name—be evaluated for local thresholding, because Candidate 2 truthfully identifies it as `TYPE=convolved_amplitude` and it uses the direct pixelwise jackknife variance in Equation (83). It must be intersected with raw-map validity and a new explicit response/support policy. `sig2noise_pixel_I` is a useful secondary diagnostic of the globally calibrated formal spatial pattern, not an interchangeable estimate. `weight_I`, `weight_formal_I`, `coverage_bool_I`, and the filtered amplitude aliases are not presently safe as confidence, gain, or direct feedback amplitude. Therefore no filtered product is yet safe for production fruit-loop support selection. Correlated filtered pixels reduce the number of independent threshold trials, inflate connected regions relative to independent-pixel intuition, couple dilation to the kernel scale, and can create edge-adjacent false support where response or fill bias is not guarded. Blank controls and injections must calibrate these effects before filtered products select production feedback.

17 Validation plan

17.1 Mathematical unit tests

Tests must implement every limiting case in Section 13, nonuniform variance, low positive weight, the zero/non-finite-weight deterministic-input precondition, full/partial/one/zero support, deterministic fill, response correction, signed-kernel support, and exact input/output taper powers. Additional matrix tests should compare the complete small-matrix AC_xA^T to diagonal and off-diagonal outputs.

17.2 Implementation and regression tests

Require sequential/OpenMP equality, kernel normalization and conditioning, formal variance/weight reciprocity, identical science/jackknife operators, support-floor invalidity, metadata truthfulness, exact $N = 2$ sample variance, mean-subtracted $N = 1$ rejection, known-zero-mean singleton behavior, and configuration rejection of unsupported fractional taper. Product-contract tests must check every extension's TYPE, BUNIT, applicable response/support status, and empirical-estimator metadata.

17.3 Same-SHA Unity validation

The authorized human should build and run the exact candidate SHA on Unity using the SSH alias `unity_toltec`; Codex must not connect directly. The generated evidence should identify source SHA, dependency/compiler identity, complete low-level YAML, input identities, dirty state, and all TolTEC arrays. The run must enable `convolve`, formal outputs, empirical noise outputs, and at least two (preferably several) realizations, and must include variable coverage plus interior and guarded-edge regions. Run the normal product inventory, log-error, provenance, configuration, and numerical comparators without required-data skips.

17.4 Blank controls

Measure pixelwise empirical variance, ratios/residuals against formal and calibrated variance, spatial correlation functions or covariance kernels, edge-distance dependence, false S/N exceedances, and false morphological support. Acceptance bounds should be chosen from the finite-sample sampling distribution, the number of effectively independent filtered modes, and repeat/control variability—not invented after viewing one result.

17.5 Injection and recovery

Inject compact, beam-shaped, resolved, and extended templates over multiple amplitudes and edge distances in all arrays. Measure stored filtered amplitude against the exact applied operator, any proposed response-corrected amplitude, uncertainty coverage, S/N calibration, centroid/morphology bias, and support selection completeness/false discovery. Pre-register acceptable deviations from analytic operator predictions and derive empirical tolerances from blank controls and repeated injections.

18 Conclusions

The mathematical authority is the estimator and its full covariance. In the diagonal approximation, formal variance is obtained only by squaring the exact signal coefficients. Response correction, support, and confidence are separate deterministic or policy transformations and cannot be folded into inverse variance without changing its meaning. The implementation verdict is completed in Section 14 after independent derivation.

A Implementation source references

All baseline references are to 9aae0e669384c5c0c0dda93debc194d6b8dac787; candidate references are to 02a198cbfb379eaf6ab279c5a3d44ee73ff90435 unless explicitly labeled Candidate 1.

Table 3: Audited source locations.

File	Lines	Evidence
include/citlali/core/ mapmaking/wiener_filter.h	947–980	FFT/circular convolution and normalized template.
same, baseline and candidate	1055–1311	Edge science/coverage masks, dilation, same-map median fill, taper/window, pre-filter signal/weight/kernel operations.
same, baseline	1314–1467	Signal convolution, thresholded h^2 variance, erroneous support division, and output window.
same, Candidate 1	1378–1458	Removal of support division but retained cov_cut input exclusion.
same, candidate stack	1378–1468	All-positive-weight h^2V propagation, relative h^2 floor, reciprocal variance, output window.
same, candidate stack	1475–1515	Fixed-window filtering of each noise realization.
include/citlali/core/ mapmaking/wiener_filter_omp.h	254–720, 837–864	OpenMP edge/filter/noise sequence and convolution counterpart.
src/citlali/core/mapmaking/ map.cpp	766–950	Welford $N - 1$ variance, global median scale, final empirical weight, two S/N definitions, direct uncertainty.
include/citlali/core/config/ post_processing_config_ validation.h	28–42	Candidate rejection of cosine taper for unit-sum convolution modes.
include/citlali/core/pipeline/ noise_weight_policy.h	7–17	Formal-versus-empirical filtered normalization policy.
include/citlali/core/pipeline/ output_policy.h	95–135, 175–180	Noise/product enablement and effective-config normalization lifecycle.

File	Lines	Evidence
<code>include/citlali/core/engine/detail/map_image_output_impl.h</code>	60–107	Filtered-output classification and response-normalization flag passed to writers.
<code>include/citlali/core/pipeline/map_image_output_helpers.h</code>	12–219	All audited FITS image producers, empirical keys, aliases, and failure checks.
<code>include/citlali/core/pipeline/fits_image_metadata_keys.h</code>	53–85, 107–221	Current scalar keys, units/types/descriptions, estimator and point-alias metadata.
<code>include/citlali/core/pipeline/fits_image_units_kernels.h</code>	5–67, 108–170	Unit construction and exact TYPE/DESCRIP strings.
<code>include/citlali/core/pipeline/fits_image_hdu_names_wcs.h</code>	5–68	Exact image extension names.
<code>validation/product_contracts.json</code>	448–476	Candidate filtered-map HDU inventory.
<code>tests/test_utils.cpp</code>	316–493	Kernel conditioning, fixed variance, partial support, low weight, $N - 1$, stability, mean-subtracted $N = 1$ rejection, known-zero-mean singleton behavior, and metadata tests.
<code>tests/test_config_scaffold.cpp</code>	3792–3811, 4608–4651	$N \geq 2$ configuration policy and taper availability tests.

The frozen consumer was inspected read-only only after derivation. At the mainline parent 9aae0e669 plus its uncommitted draft, `include/citlali/core/timestream/timestream.h`, lines 2161–2250, validates filtered selector HDUs and fails closed if the corrected contract is absent; lines 2664–2746 combine pixel/direct S/N, `coverage_bool_I`, raw-map validity, morphology, and raw feedback amplitude. This consumer is evidence of intended use, not authority for the equations.

B Product and test matrix

In Table 4, “candidate pass” means only that the value is coherent with the conditional stored-amplitude diagonal contract; it does not waive the fill, response, covariance, or external-validation gaps. The common map unit in the accepted fixture is `mJy/beam`.

Table 4: Current product-by-product contract and audit disposition.

Extension or concept	Value and estimator	Current metadata and producer	Disposition
<code>signal_I</code>	Equation (67); stored fixed convolved amplitude, circular operator, output zero outside T ; map unit.	UNIT/BUNIT=mJy/beam, generic DESCRIP; helper 19–22, filter 1055–1468. No TYPE, response template, fill, support, or covariance policy.	Baseline and candidate same signal. Add fixed-convolved-amplitude TYPE, FILTEROP=unit_sum_convolution, circular boundary, fill/window policy, and explicit feedback withholding. Never direct fruit-loop feedback under the current design.
<code>weight_formal_I</code>	Candidate Equation (76); inverse conditional diagonal variance of stored amplitude; map-unit ⁻² . It snapshots the weight entering filtered empirical calibration and can therefore inherit an upstream scalar calibration. Written when empirical products are expected.	TYPE=formal, BUNIT, generic formal description; helper 62–67.	Baseline fails Equations (72); candidate passes conditionally. Add estimator ID, input-weight provenance, and conditioning declarations. Not confidence, support, or gain.
<code>noise_variance_I</code>	Equation (57); direct per-pixel sample variance of identically filtered conditional amplitudes; map-unit ² .	BUNIT/DESCRIP, NNOISE, VAREST=central_sample_variance, VARDDOF=1, MEANSUB=true; helper 68–80, map 766–870.	Baseline central divisor is wrong; candidate passes $N - 1$ /stability. State conditional fixed-mask/fill and diagonal-only. Closest uncertainty for local selection after validation.
<code>weight_I</code>	Formal weight, or Equation (79) when normalization applies; not generally $1/s_i^2$; map-unit ⁻² .	TYPE=formal or empirical; empirical case adds N keys, EMP_SCALE, WVARMED; helper 24–50, map 872–940.	Describe as a globally jackknife-calibrated formal pattern and state its calibration region. Candidate algebra coherent. Unsafe as support, confidence, or gain.

Extension or concept	Value and estimator	Current metadata and producer	Disposition
<code>sig2noise_I</code>	Exact legacy alias of Equation (81); dimensionless.	TYPE= <code>pixel</code> , N keys, legacy-alias DESCRIP; helper 143–159.	Candidate coherent but name ambiguous. Retain only as deprecated alias. Not primary fruit-loop selector.
<code>sig2noise_pixel_I</code>	Equation (81); global-calibrated formal-pattern standardized amplitude; dimensionless.	TYPE= <code>pixel</code> , N keys; helper 153–159.	Does not equal direct-jackknife S/N generally. Useful secondary diagnostic after validation, not confidence or feedback gain.
<code>point_source_flux_I</code>	Exact alias of <code>signal_I</code> ; no response correction for convolve; map unit.	Candidate TYPE= <code>convolved_amplitude</code> , no RESPNORM; helper 186–196. Baseline point-source description plus RESPNORM=1.	Baseline metadata definite error; candidate truthful compatibility alias. Prefer a non-photometric name; never treat as mJy integrated flux or direct feedback amplitude.
<code>point_source_uncertainty_I</code>	Equation (82); uncertainty of the same stored convolved amplitude; map unit.	Candidate TYPE= <code>convolved_amplitude</code> plus all N keys; helper 198–207.	Candidate truthful, conditional, diagonal. Not uncertainty of response-corrected photometry. May support selection only with response/support gates.
<code>sig2noise_point_source_I</code>	Equation (83); stored amplitude divided by direct jackknife uncertainty; dimensionless.	Candidate TYPE= <code>convolved_amplitude</code> , N keys, no RESPNORM; helper 209–217.	Legacy name misleading, but ratio is coherent. Closest local-threshold product after blank/injection validation and explicit response/support intersection; currently fail-closed.
<code>coverage_I</code>	Original effective integration time; seconds; not filtered response.	UNIT/BUNIT= <code>sec</code> , DESCRIP; helper 116–120.	Authoritative exposure diagnostic. Do not use as inverse variance or edge response.

Extension or concept	Value and estimator	Current metadata and producer	Disposition
coverage_ bool_I	$\mathbf{1}[\neg(W_i < \tau_{\text{cov}})]$ from final <code>weight_I</code> ; dimensionless binary display mask, with threshold found using <code>cov_cut</code> . This differs from $\mathbf{1}(W_i \geq \tau)$ for non-finite values.	BUNIT=N/A, WTTHRESH, support-mask DESCRIP; helper 121–132.	Legacy final-weight comparison mask, not complete validity, the stored edge window, $q^{(k)}$, or response. At zero threshold, zero and NaN weights can be marked one. Insufficient for fruit-loop edge support.
kernel_I	Convolved mapmaking kernel image, not necessarily the filter template h or a response plane; repository currently assigns map unit.	TYPE= raw kernel type, BUNIT, FWHM, DESCRIP; helper 84–103.	Add identity of both input kernel and applied convolution. Do not infer point-source or absolute support without template conventions.
edge window T	Binary science-plus-dilated-guard window in memory; Equation (68).	No FITS image extension in the inspected producer or product contract; scalar diagnostic state records thresholds/counts/radii/background.	Persist a named validity/window policy or sufficient reproducible provenance. Binary validity is not inverse variance.
h^2 validity support	Equation (75); used only against 10^{-6} .	Not stored and no image metadata.	Correct numerical floor mechanism, but not an adequate exposed geometric support contract.
response, absolute support, taper, confidence	No corresponding convolve FITS planes. Accepted taper is binary.	Absent from helpers and product contract.	Keep production selection withheld. A future product must name its template or support norm, eligibility, and floor; keep confidence/gain separately named.
empirical scale	Scalar g and median ratio ρ in Equations (77)–(78).	EMP_SCALE and WVARMED on empirically calibrated <code>weight_I</code> .	Correctly scalar, but add calibration-region/policy identity and usable-pixel count. Never feedback gain.

All image products also require producer SHA/version, observation/grouping, array and wavelength, Stokes and iteration, effective filter type, applied kernel identity, boundary convention, and response, support, and fill policy at a level sufficient to reproduce the estimator. Existing

FITS/WCS and product provenance provide some of these identities, but none currently expresses the complete convolve statistical contract.

Table 5: Equation-to-test and validation matrix.

Contract	Existing candidate evidence	Required additional gate
Unit-sum interior, Equations (19)–(21)	<code>unit_sum_convolution_preserves_well_conditioned_behavior;</code> <code>convolve_propagates_uniform_diagonal_variance.</code>	Fixture-runtime $\sum h^2$ assertion for each array and independent small-matrix covariance test.
Fixed partial support, Equations (26)–(30)	<code>convolve_does_not_renormalize_variance_by_valid_support.</code>	Explicit signal, fill, output-mask, one-support, zero-support, and off-diagonal covariance cases; sequential/OpenMP equality.
Low positive weight, Equations (8) and (74)	<code>convolve_variance_includes_all_positive_weight_inputs.</code>	Test nonuniform signal and variance together, including guardband input below <code>cov_cut</code> .
Median fill, Equations (14)–(15)	None.	Bootstrap/analytic or injection evidence bounding omitted median-fill variance/cross-covariance; record conditionality.
Taper, Equations (41)–(43)	Configuration rejects cosine for convolve and lowpass-only.	Keep rejection; add a dormant-path algebra regression if the code remains reachable, or redesign before enabling.
Empirical variance, Equation (57)	Exact $N = 2$, large-offset stability, mean-subtracted $N = 1$ rejection, known-zero-mean singleton behavior, and metadata tests.	Same-SHA multi-realization runs and direct numerical comparison to an independent sample-variance calculation.
Global calibration, Equations (77)–(79)	Focused helper/map tests and reported regression suite.	Test exact median region, empty/invalid region, scale identity, and prove <code>weight_I</code> metadata/product inventory on real output.
Response, Equations (47)–(49)	Candidate removes false convolve <code>RESPNORM</code> and changes <code>TYPE</code> .	Delta, beam, compact, resolved, and extended injection/recovery at interior and multiple edge distances.
Support, Equations (35)–(36)	Internal h^2 floor only.	Decide product/floor from blank and injection evidence; signed-kernel tests; verify edge false support.
Filtered covariance, Equation (12)	No covariance product.	Blank-map spatial correlation/covariance by array and coverage/edge bin; calibrate threshold trials, component sizes, and dilation.
FITS contract	Candidate metadata/unit tests and <code>product_contracts.json</code> .	Same-SHA all-array product inventory, exact metadata, no unexpected error-level logs, and no skipped required data.

C Independent-derivation record

Sections 4–13, the recommended estimator in Section 15, and the pre-audit fruit-loop consequences were drafted on branch `codex/convolve-contract-audit` from mainline `9aae0e669384c5c0c0dda93debc194d6b8dac787` before inspecting the diffs or source contents of candidate commits `fa8f8a5b9` and `02a198cbf`. Candidate code is treated only as evidence against the numbered equations.